The Lexicon Is the Load-Bearing Wall: How Schema Term Quality Decides Entity-Layer Fate in Dictionary-Driven Enrichment at Production Scale

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Abstract

Dictionary-driven mention extraction is a common design in production knowledge-base pipelines: it is cheap, reproducible, and idempotent. We report a production incident in which this design, fed by a machine-generated schema, silently collapsed the entity layer of a 61,137-document scholarly knowledge base into **575 entities** carrying **19.2 million entity links**—every entity a bare type word such as "method" or "energy". No conventional monitor fired: enrichment jobs completed, throughput was normal, and link volume looked like growth. We trace the collapse through three layers—the extraction mechanism (pure lexicon scan), the lexicon's provenance (a batch-generated schema whose per-type aliases form a category taxonomy rather than a domain terminology), and the healthy counter-example in the same deployment (78 documents, a hand-curated 3,742-term lexicon, 1,608 well-formed entities). We then describe a three-source, anti-hallucination lexicon generation method (transplanted seed terms, corpus-candidate classification, and per-type expansion, all gated by corpus attestation; 3,702 and 1,279 terms for the two affected schemas) and a full-chain rebuild over 766,501 re-enqueued jobs that doubled as a load test, surfacing and fixing five pipeline pathologies (a whole-corpus join in graph projection that saturated a 112-core database at load 500; two link-insertion deadlock classes; an $O(N^2)$ acronym check amplified by the richer lexicon itself—137× link throughput after the fixes; and a

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parameter-order bug whose rows were silently swallowed by `INSERT IGNORE` for seven hours while every job-level monitor read green). The rebuilt layer holds **1,871 canonical entities with 25.1M links and 44k typed relations** on the 61,137-document library—3.25× the collapsed cardinality, a relation layer where none functioned, and scoped retrieval coverage up +6.6/+7.3 points over the repair window. Two audits bound the claim: a single-variable channel ablation shows the retrieval gain does *not* run through entity-channel fusion votes (metrics bitwise identical with every entity-dependent channel off), and an LLM-assisted audit of a seeded 100-link sample puts rebuilt link precision at **0.85** (95% CI [0.77, 0.91]; cross-model-confirmed at 96% agreement on the parseable subset)—1.00 on domain terminology (n=25), 0.80 on the generic tail that survives repair, a documented boundary rather than a triumph. We argue the incident generalizes: wherever an LLM drafts a schema whose aliases then become the extraction surface, lexicon quality is load-bearing infrastructure, and it needs its own observability—scale-invariant link-concentration and entity-name-distribution monitors that standard pipeline dashboards do not provide by default.

Keywords: knowledge base construction, schema quality, terminology extraction, entity linking, production incident, observability

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1. Introduction

Large scholarly knowledge platforms increasingly generate their subject schemas with LLM assistance: a batch tool drafts entity types with a handful of aliases each, a validator checks structural well-formedness, and the schema is bound to knowledge bases as configuration (our own batch generator is the first villain of this story; LLM-for-ontology generation is surveyed in [5]). This is attractive at scale—our deployment covers 80,855 parsed journal articles (80,852 active, 7,229,225 text chunks) across two large knowledge bases and several smaller ones—and it keeps humans out of a loop that would otherwise require domain lexicographers per discipline.

This paper is about what that design decision does to the entity layer when the enrichment pipeline extracts mentions by dictionary scan. In our system, stage-K1 mention extraction is deliberately LLM-free: a schema's `entity_types` main names and aliases constitute the scan dictionary. The choice is defensible—dictionary scan is cheap enough to run over 7.2 million

chunks, deterministic, idempotent under re-ingest—and it performed adequately on the knowledge base whose schema had been hand-evolved over years. It failed catastrophically, and silently, on the two knowledge bases whose schemas came out of the batch generator.

The failure signature: a 61,137-document nuclear-electronics knowledge base (104112E6) accumulated **19.2 million entity links onto 575 canonical entities**; the entities were not domain concepts but type words—"method", "parameter", "energy". A 13,408-document physics knowledge base (693372D5) showed the same disease at 2,018 entities. Meanwhile every operational signal was green: enrichment jobs completed normally, link volume read as growth, and retrieval endpoints answered. The entity graph had degenerated into a bipartite "type-to-document" index while presenting, to every surface-level check, as a healthy knowledge graph.

We contribute:

1. **A forensic record of entity collapse induced by schema-lexicon quality**, including the three-layer causal chain (Section 3) and an analysis of why standard monitoring is blind to it—the observables that finally expose it are the scale-invariant concentration of links on the top entities and the name distribution of top-linked entities, with entity cardinality as a corroborating (and manipulable-in-both-directions) secondary signal (Section 7).
2. **A three-source, corpus-attested lexicon generation method** (Section 4) that repairs the affected schemas without manual lexicography: 2,284 transplanted seed terms from a healthy near-domain library, 7,391 corpus-frequency candidates classified by LLM, and bounded per-type expansion—yielding 3,702 real terms for nuclear electronics (v2.1) and 1,279 for physics (v1.1), with 223 legacy cross-type conflicts repaired in passing.
3. **A full-chain rebuild at production scale** (Section 5): vertex- and table-scoped teardown across graph store, relational registry, and search projection; 766,501 re-enqueued jobs; and the pipeline pathologies the rebuild stress test surfaced, each with a measured fix—most dramatically a 137× link-throughput recovery once an $O(N^2)$ alias check, itself amplified by the new richer lexicon, was cached.
4. **Before/after evidence and observability recommendations** (Sections 6–7): entity cardinality 575→1,871 with a functioning relation layer (44k typed relations, 1.56M evidence rows); scoped

retrieval trajectory +6.6/+7.3 cov@10 with a channel-level attribution bound (Section 6) and an audited link precision of 0.85 with a 1.00 domain stratum; and four monitors—led by the scale-invariant top-10 link concentration—that would have caught the incident on day one.

The larger claim we defend: in dictionary-driven pipelines, **the schema lexicon is not descriptive metadata but operative extraction surface**. Treating it as configuration-file decoration—easily batch-generated, rarely audited—is a structural mistake that this incident renders measurable.

2. Background: dictionary-driven enrichment

Our enrichment pipeline (four stages per document: mentions → links → relations → graph) extracts mentions in stage K1 by normalized dictionary scan: every schema alias match over chunk text yields a `raw_mention` with an entity-type hint; stage K2 links mentions to canonical entities by normalized form + type; K3 lifts selected mention pairs into typed relations with an LLM pass; K4 projects the per-document entity subgraph into the graph store and search-side entity index. Retrieval consumes all four: an entity channel seeds personalized PageRank, a co-occurrence channel, and entity-vector similarity all trace back to linked mentions.

Why no LLM in K1? Cost and determinism. At 7.2M chunks, a per-chunk LLM extraction pass is a material recurring expense and a re-ingest hazard (non-determinism breaks idempotent rebuild); a scan is $O(\text{corpus})$ with zero marginal model cost. The design bets everything on one input: the dictionary. That bet is invisible in the architecture diagram and total in its consequences—a point Section 3 makes empirical.

The healthy control in our own deployment: a hydrogeology knowledge base whose schema was hand-evolved to 3,742 genuine domain terms (78 documents → 1,608 canonical entities, well-distributed). The contrast cases: schemas produced by our batch tool, whose output per entity type is a small set of *category* aliases (4–8 per type)—a taxonomy of the subject, not a terminology of it (the distinction between terminological and taxonomic vocabulary is classical [6], and the adjacent question of how to measure knowledge-graph quality is itself an established literature [3]).

Neither half of our pipeline is novel in isolation. Dictionary- or gazetteer-driven entity recognition is one of the oldest NER traditions [2],

Table 1: The collapse, measured.

KB	Docs	Entities	Links	Schema provenance
104112E6 (nuclear electronics)	61,137	**575**	~**19.2M**	batch-generated
693372D5 (physics)	13,408	**2,018**	$\approx 10^6$ (est.)	batch-generated
hydrogeology (control)	78	1,608	normal	hand-evolved

and corpus-statistical term recognition with linguistic evidence goes back to C-value/NC-value [1]—our corpus-candidate classification source (Section 4) is that automatic-term-recognition program reborn with an LLM in the classifier seat. Likewise, detecting and repairing degraded knowledge-graph content is an established concern [3], and production data-validation systems routinely gate model inputs with schema- and statistics-level checks [4]. What is missing from that prior art, and what this incident supplies, is the specific failure locus: when the dictionary *is* the extraction model, a structurally valid but terminologically hollow schema passes every existing gate—term-recognition research assumes the candidate pool is real corpus vocabulary, and data-validation systems check schema shape and distribution drift, not whether a lexicon's entries are domain-specific. The monitors we propose (Section 7) close exactly that gap.

3. The collapse, forensically

The observation. During an unrelated storage investigation (2026-10-08), a routine entity-quality sample on 104112E6 returned type words where domain concepts were expected. The full picture (Table 1):

61,137 documents carrying ∼19.2M links is ∼315 links per document on average, and ∼33,000 links per entity: every document "mentions" the type vocabulary constantly. The graph is a bipartite type–document index wearing a knowledge graph's clothes.

The incident timeline (dates stated plainly, because a repair of this scale compressed into a two-day window is exactly the kind of claim a reader should be invited to check):

The compression is real and we do not smooth it over: the tooling was written under incident pressure on day one, and the evaluation experiments are retrospective audits of a settled production state, not a pre-registered protocol—a limitation Section 8 keeps visible.

Table 2: Incident timeline.

Date	Event
2026-10-08	Collapse discovered (storage investigation); three-layer root cause traced same day; three-source lexicon generator built, run, and committed (nuclear v2.1, physics v1.1); rebuild driver executed; 766,501-job re-enqueue begins against the live fleet
2026-10-08 (night)	Silent-swallow incident: 7 hours of zero link-registry growth under all-green job monitors; caught by row-count reconciliation; 51,459+24,372 empty jobs reset; recovery measured at +611k rows/min
2026-10-09	Four-stage drain completes on both libraries; per-KB statistics recomputed; entity vectors reprojected; the two smaller diseased libraries (1,324+119 docs) treated with the same playbook
2026-10-09→10	Post-hoc audits on the settled system: scoped trajectory refresh, single-variable channel ablation, seeded 100-link precision audit, fleet-wide concentration measurements (the RAND seed 20261009 records the audit date); manuscript drafted and revised

Layer 1 — mechanism. K1 scans for schema aliases. When the alias set for type `method` contains common words ("method", "approach", and their CJK equivalents), every occurrence of those words in 7.2M chunks becomes a mention, and K2's normalized-form+type linkage folds them onto one canonical entity per (form, type) pair. The mechanism executed flawlessly on garbage input.

Layer 2 — provenance. The dictionaries were the schemas. The batch generator, prompted to "draft entity types with aliases for subject X", produced exactly that: type-level category vocabulary. Nothing in its validator checks whether aliases are *domain-specific*—structural validity is the gate, and category words are structurally valid.

Layer 3 — the control. Hydrogeology's 3,742 hand-evolved terms over 78 documents yield 1,608 entities ≈20.6 per document of genuine concepts. Collapse correlates with lexicon nature, not corpus size or pipeline version.

Why nothing fired. Job-level observability was all green (jobs DONE, throughput nominal, error rate zero). Link volume—if anything—read as

success (19.2M links!). The observables that expose the collapse are (a) the concentration of links on the top entities—the collapsed state measures 0.419 with its top-10 entities being pure type vocabulary, versus 0.18 for the healthy control (Section 7's monitor (a) gives the measurement and its provenance)—and (b) the name distribution of top-linked entities, which is pure type vocabulary. Entity cardinality per document—two orders of magnitude below the control—corroborates, but cardinality is a coarse signal gameable in both directions (the small-library treatment of Section 8 shows category-word aliases inflating it), which is why it is the secondary, not the primary, monitor. Neither observable is provided by standard pipeline dashboards by default. This blindness generalizes to any dictionary-driven system whose dashboards count jobs and rows.

4. Lexicon generation, once and correctly

The repair requirement: produce genuine domain terminology at scale, without per-subject lexicographers, without hallucinated terms—and make it a repeatable tool, not a one-off patch. Our generator implements three sources, all gated:

1. **Seed transplant (2,284 terms).** A healthy near-domain library (the RADAI corpus knowledge base, same academic register) donates its terms with types. Transplant is the highest-precision source; corpus attestation below still applies.
2. **Corpus-candidate classification (7,391 candidates).** High-frequency n-grams from the target corpus's own text are proposed—concretely, the top 8,000 n-grams (unigrams and bigrams) by chunk frequency over a 3,000-chunk-per-KB sample, keeping candidates with frequency ≥ 5 after a boilerplate and stop-word filter (section vocabulary such as *acknowledgements/approach/using*; closed-class English words). An LLM classifies each surviving candidate into a schema entity type or rejects it. The corpus floor is the anti-hallucination gate: a term the corpus never attests is not admitted, whatever the LLM thinks of it.
3. **Per-type expansion ($\leq$60 per type).** For each entity type the LLM proposes additional domain terms, given the type description and the transplanted seeds as in-context anchors; the same corpus-attestation gate applies. The bound keeps any single type from flooding.

Generation parameters, for reproducibility: the classifier/expander is an AWQ 4-bit build of the Qwen3.6-35B-A3B checkpoint (community build `cyankiwi/Qwen3.6-35B-A3B-AWQ-4bit`; we retain the deployment string verbatim since the endpoint serves it under a historical alias), served via vLLM behind a round-robin router, temperature 0.2, $\leq$4,000 output tokens per call, three retries on empty or non-JSON responses; the entire three-source run for both schemas is a single idempotent invocation of the generator.

Quality gates beyond attestation: *catch-all types* (`term`, `term_cn`, `model_or_code`, `concept`, `question`) accept no classified noise—they are the natural sink of classification uncertainty and the main entry route for taxonomy garbage; a boilerplate filter removes layout/section vocabulary; cross-type alias conflicts resolve by a specificity ordering (the more specific type wins); and the schema validator runs as a final structural gate before any version bump. The same pass repaired 223 legacy cross-type conflicts in existing schemas.

Output: **nuclear_electronics v2.1 = 3,702 terms; physics v1.1 = 1,279 terms**. 104112E6 rebinds to nuclear_electronics (its corpus is nuclear-domain journals). Both versions are idempotent re-runnable—the tool is now part of the schema lifecycle, not an incident artifact.

An honest note on the asymmetry (physics 1,279 versus nuclear 3,702): the physics corpus is smaller and its seed transplant thinner; v1.1 is a floor, not a ceiling, and the runbook treats small-physics libraries as follow-up work (Section 8).

5. Rebuild at scale

Repairing live knowledge bases requires tearing down the collapsed entity layer without stopping ingestion or retrieval. The rebuild driver executes per knowledge base: delete graph-store vertices with edges; delete the four relational tables (mentions, links, relations, graph-projection queue rows) in bounded chunks; delete the search-side entity projection by query; then re-enqueue all documents through the four stages—**766,501 jobs** across the two affected libraries. The rebuild ran against the production fleet (five multi-worker enrich instances, one links-dedicated consumer per KB) while normal retrieval served.

The rebuild doubled as the system's first full-corpus stress test of the *rich-lexicon* configuration, and four pathologies surfaced—each now a fixed,

committed, measured lesson:

- **B90 — unbounded projection join.** Graph projection originally joined each document against *whole-corpus* entities and the *whole-corpus* link table; at 80k documents and millions of links a single projection took 25+ minutes, and 80 concurrent consumers saturated a 112-core MySQL (107 cores, load 500). Scoping the join to the document's own mentions (an index-ordered path) is semantically equivalent—an entity projects through documents that mention it—and took projection to seconds, database to 29 cores.

- **B91/B93 — link-insertion deadlocks.** Two deadlock classes in link insertion: unordered INSERT-SELECT acquiring gap locks in interleaved order (fixed by pinning insertion order to `mention_id`), and per-action round-trip inserts (fixed by folding all actions into a single UNION-ALL derived-table INSERT-SELECT). Deadlock error 1213 went to zero.

- **B94 — the $O(N^2)$ the new lexicon amplified.** Alias canonicalization checked every acronym pair quadratic in alias count—95 of 101 seconds per links task, burning the GIL while 16 threads queued. Caching canonical forms by schema signature (lexicon version + alias count) took it to 0.05 s. Net link throughput: **8/min → 1,100/min (137×)**—the collapse-era lexicon was so small it hid this bug for months; the *repair* exposed it in an afternoon.

The last point deserves emphasis in prose, not just as a bullet: the collapsed lexicon (575 de-facto terms) masked an $O(N^2)$ hot path because N was tiny. Correcting the lexicon did not merely improve extraction—it changed the computational regime of the pipeline and surfaced latent engineering debt. Lexicon quality is load-bearing in the most literal sense.

The rebuild also supplied its own fifth pathology, and it is the one that most cleanly demonstrates our observability thesis. The B93 single-statement link writer shipped with a parameter-order bug: the SQL's first placeholder was the knowledge-base id (`UNHEX(%s)`), but the call bound the term form there first—so every row carried `UNHEX('alloy')` = NULL as its kb_id, violating the NOT NULL column, and **INSERT IGNORE silently swallowed the entire insert**. Jobs completed. Entities were created. Throughput read 1,100 jobs per minute. Only one observable was wrong: the link

Table 3: Before/after, entity layer.

Metric	Collapsed	Rebuilt	Control (hydro)
KB1 documents	61,137	61,137	78
KB1 canonical entities	**575**	**1,871** (3.25×)	1,608
KB1 entity links	∼19.2M (type words)	**25,097,704** (terms)	normal
KB2 entities / links	2,018 / $\approx 10^6$ (est.)	**2,730 / 5,009,755**	—
KB1 typed relations	∼0 effective	**44,043** + 1.56M evid. rows	—

registry stopped growing—for seven hours. We detected it via row-count reconciliation against job completions, fixed the binding, reset 51,459 + 24,372 silently-empty jobs, and measured recovery at +611k rows in the first 60 seconds. Job-level DONE is not row-level truth; any bulk writer guarded by INSERT IGNORE needs an affected-rows reconciliation, and pipeline dashboards need a target-table growth rate beside the job queue.

Rebuild final state. All four stages drained across both knowledge bases: mentions, links, relations, graph projection. The repaired entity layer, measured: **KB1 1,871 entities carrying 25,097,704 links (from 575 / ∼19.2M collapsed), 44,043 relations with 1.56M evidence rows (from effectively zero); KB2 2,730 / 5,009,755 / 12,774.** Spot-checked canonical entities (germanium detector, hpge detector, neutron, preamplifier, scintillation detector) all project correctly into the graph store. Retrieval on KB1 answers in 0.6–1.3 s warm with all channels active.

6. Before/after

Entity layer. The headline repair (all numbers from the production database, reproducible SQL in the released evidence pack; Table 3):

The relation layer did not exist in any usable form before the repair; it now carries schema-typed, LLM-adjudicated relations with per-relation evidence quotes, and is what the retrieval graph channel traverses.

One reading note on absolute counts: link totals *rise* after repair in both libraries (KB1 ∼19.2M→25.1M; KB2 $\approx 10^6$ →5.0M)—the repaired lexicons carry thousands of real terms that match more often than the category-word dictionaries did in these corpora. This is expected, and it is precisely why

count was never the quality signal: the distinguishing observables are how the links distribute across entities (concentration) and onto which names (top-entity audit), not how many there are.

Name quality, honestly. The top-linked entities after repair are `be` (2.26M links), `alpha`, `gamma`, `beta`, `method`—generic words and short ambiguous symbols still dominate the link head. The corpus-attestation gate can only guarantee that a term *occurs*; it cannot distinguish a high-frequency common word from a domain term. The `be` case is instructive because it is not even a ranking problem but a normalization one: the element symbol *Be* and the English auxiliary verb *be* are distinct in the source text, but K1's scan normalizes case (ASCII terms match case-insensitively with word boundaries), so every auxiliary "be" in 7.2M chunks links to the beryllium entity. A case-sensitive allowlist for short chemical symbols plus an English stop-word exclusion at admission would remove the bulk of this mass; `alpha`/`gamma`/`beta` are harder—genuine nuclear-spectrum symbols that also serve as generic Greek-letter variables—and need context-type constraints, not stop-word lists. Genuine terminology (hpge detector, neutron, preamplifier, …) is present and correctly linked and projected—but the repair relocated the lexicon-quality problem from "no terminology at all" to "terminology plus a generic tail." We consider this the honest headline of the incident: repairing a collapsed lexicon is one battle; lexicon hygiene is a permanent operational discipline (Section 7's monitors exist precisely for this).

Retrieval trajectory, with a channel-level attribution bound. On the scoped evaluation slices (same runner, same corpus scope) the system moved from the collapsed-era run (2026-09-24) to the rebuilt state (2026-10-09): associative queries cov@10 0.533→**0.599** (+6.6pt, n=48), single-intent queries 0.603→**0.675** (+7.3pt, n=39) with MRR −2.0/+5.0pt respectively. The two-week window also carried retrieval-configuration evolution (document-fusion re-ranking, bilingual alias expansion, kNN search-depth tuning), so the trajectory measures system evolution *including* the lexicon repair. To bound the attribution we ran a single-variable channel ablation under the identical post-rebuild configuration (in-process planner, same corpus scope, same questions): a *no-entity* arm with every entity-dependent channel disabled (co-occurrence, graph, entity-vector, PPR) against the full configuration. Result: **metrics are bitwise identical on both families**—cov@10/R@10/MRR unchanged (nea 0.599/0.938/0.735, ner 0.670/0.923/0.770; the only movement anywhere is nea cov@20 −1.0pt, 95% CI [−3.1, 0.0])—while p50 latency falls from 1.46 s to 0.52 s. The

co-occurrence channel fired on only 6/48 association queries (its adaptive gate suppresses it when base retrieval already serves co-mention documents) and changed no outcome.

Channel-liveness check (the ablation's alternative explanation, closed). A bitwise-identical result admits a defensible alternative reading—that the entity channels are not wired, or that their gates never open—so we ran a forced-trigger check on the 99-question natural-phrasing set (template words banned), the distribution the co-occurrence gate was designed for: there the channel fired on **54/99 queries** (versus 6/48 on the scoped template set), killing it changed the top-10 on 26 queries, and the paired gold-hit comparison reads **20 losses / 0 wins / 34 ties** for the off-arm—removing the channel never helps and hurts on a fifth of the set. The channel is alive, the gate opens exactly where designed, and the scoped zero-effect is the gate closing because base retrieval already serves those co-mention documents—not a dead wire.

The attribution this forces is precise and we state it against interest: the +6.6/+7.3pt trajectory is **not** carried by entity channels casting fusion votes. The entity layer's retrieval-facing value on these benchmark families runs through the lexical channel's bilingual alias expansion (entity names as BM25 terms—a mechanism that needs a real terminology to expand, and which the collapsed lexicon could not feed) and through structural surfaces (reranker context, knowledge-chain evidence for generation) documented in the companion system manuscript [7]; at the fusion layer itself, entity channels are currently a latency cost with zero measured recall benefit on these sets. The lexicon repair was necessary for every entity-consuming surface; the scoped trajectory number alone cannot be credited to it.

Link precision audit (LLM-assisted). Distributional monitors say nothing about whether individual links are *correct*. We audited a seeded, reproducible uniform sample of 100 links from the rebuilt KB1 registry (`ORDER BY RAND(20261009)`): for each link the mention span was programmatically verified against the chunk text (span integrity 100/100—offsets are exact), and an LLM judge (Qwen3.6-35B-A3B, temperature 0) received the ±250-character context with the span marked and answered a strict two-part rubric: (i) does the span denote the same concept as the entity's canonical name (different sense, different concept, or non-term mention ⇒ invalid; debatable ⇒ invalid + borderline); (ii) is the canonical name a domain-specific technical concept or a generic academic word. Results: **overall precision 0.85** (Wilson 95% CI [0.77, 0.91]); the domain stratum is **25/25 = 1.00** (CI

[0.87, 1.00]) and the generic stratum 60/75 = **0.80** (CI [0.70, 0.88]). Every adjudication, span, and reason string is released. The reading is the name-quality headline quantified: where the lexicon carries real terminology the dictionary-scan link layer is essentially perfect, and the residual error mass sits almost entirely in the generic tail. The audit is LLM-assisted rather than human-adjudicated—a limitation we keep visible (Section 8)—but its stratum split is exactly what the monitor design predicts, and the programmatic span check removes the cheapest class of audit noise.

Cross-model check (the judge-independence concern, addressed). The audit judge and the lexicon classifier are the same model family—a self-grading risk we take seriously. We re-adjudicated the identical 100 links with a second judge from a different model family and provider (Qwen3.8-Flash via an external gateway; temperature 0, identical rubric). The second judge failed to emit parseable verdicts on 45/100—a flash-tier instruction-following failure we report as-is, not hide; on the 55 it did adjudicate, verdict agreement with the primary judge is **53/55 = 96.4%** (50 both-valid, 3 both-invalid, 2 discordant—both discordances on the borderline generic word "term"), and the domain stratum agrees **14/14**. Where an independent model opined, it confirmed the primary adjudication; residual circularity exposure is confined to the unparseable subset and to the small domain stratum (n=25 primary, 14 cross-parsed). A human-adjudicated stratified sample remains the decisive control and is listed as such in Section 8.

7. Discussion

Observability for lexicon quality. Four cheap monitors would have caught both incidents at first symptom, in the spirit of production data-validation systems [4]:

(a) *Top-10 link concentration (primary, scale-invariant).* The fraction of all entity links held by the ten most-linked entities. As a share it is comparable across corpus sizes and disciplines—exactly what raw cardinality is not. Measured across the thirteen libraries of our deployment, the metric spans 0.16–0.77: the healthy and repaired large libraries run at 0.16–0.19 (rebuilt physics KB2 0.162, hydrogeology control 0.184); the rebuilt nuclear library sits at 0.345 with its generic tail; and the two small treated libraries **currently** read 0.398 (1,324 docs, 629 entities) and 0.768 (119 docs, 153 entities)—values we report as they

are, not smoothed: the smallest library also illustrates the metric's floor effect (with $O(10^2)$ entities, ten of them can hold most links under an ordinary Zipfian distribution alone), so the band should be read jointly with library size. The collapsed state itself is now measured, not gestured at: recovering the pre-incident cold database backup (2026-10-06, two days before discovery, with the collapsed enrichment still writing—4.8M links on KB1 at that moment versus 19.2M at discovery) gives **0.419** for collapsed KB1 (top-10 entities: *time, model, method, energy, temperature, field, term, tau, phase, material*—431k links on "time" alone) and **0.290** for collapsed KB2. The repair moves KB1 0.419→0.345 and KB2 0.290→0.162—direction correct, and the residual gap is the generic tail. One honesty note: the metric partly improves simply because the entity population grows (503→1,871 link-bearing entities); the name audit (c) remains the sharper discriminator of the collapse itself. We propose a band alarm (>0.35 investigation, >0.5 escalation) with an explicit caveat: the thresholds are calibrated post-hoc on this deployment's thirteen libraries, not on an external population—a starting point for adopters, not a validated norm.

(b) *Entity cardinality ratio (coarse secondary).* Canonical entities per ingested document, alarmed against a per-domain band (our control suggests O(10–20) for scholarly corpora; collapse ran at 0.009). Cardinality alone is gameable in the other direction—the small-KB treatment showed category-word aliases *inflating* the count (Section 8)—which is why it backs up (a) rather than leading. The O(10–20) band itself comes from a single hand-evolved control library and is offered as illustration, not a calibrated norm.

(c) *Top-entity name audit.* The top-K entities by link count should be domain concepts; a manual or LLM spot check of that list is a five-minute quarterly control, and the link-precision rubric of Section 6 is a ready-made protocol for a deeper sample.

(d) *Target-table growth reconciliation.* Completed-job counts reconciled against downstream row growth, mandatory wherever a bulk writer is guarded by INSERT IGNORE (the silent-swallow incident of Section 5 was invisible to job metrics for seven hours and exposed by exactly this reconciliation).

None of the four requires ground truth. The small-library treatment (Section 8) doubles as the monitors' out-of-sample validation: after the same repair playbook, entity counts contracted (870→629, 209→153—the cardinality signal normalizes) while the concentration monitor still flags both libraries—one above the investigation threshold ($0.398 > 0.35$), one above the escalation threshold ($0.768 > 0.5$)—correctly, since their tails remain generic—demonstrating that the two monitors separate "cardinality repaired" from "distribution still skewed", which cardinality alone cannot.

LLM extraction versus dictionary scan. The incident is not an argument for LLM-everywhere. It is an argument for honoring the input contract of cheap deterministic components: if extraction is a scan, the dictionary is the model—and it deserves the same evaluation discipline (held-out attestation, distributional checks) as a trained model. Batch-generating that dictionary with an LLM and shipping it on structural validation alone is the actual failure mode.

Schema tooling implications. The batch schema generator produced category taxonomy where a terminology was needed; the repair gave the schema a real lexicon (3,702 terms) and folded lexicon generation into the schema lifecycle as an idempotent tool. What remains open—and what the rebuilt top-entity distribution makes plain—is that *admission* criteria (corpus attestation) are necessary but not sufficient: a lexicon also needs *ranking* criteria that demote high-frequency generic words and short ambiguous symbols (IDF-style corpus contrast, bilingual alias consistency, length/type priors). The observation that the collapsed state and the repaired state differ by a factor of 3.25 in entity cardinality—while both pass structural validation—is the strongest argument we have for treating lexicon quality as first-class schema infrastructure with its own acceptance gates, not as a one-time repair.

8. Limitations and honesty

1. **The generic tail survives the repair**: top-linked entities remain short generic symbols; admission gates without ranking gates move the problem, they do not remove it.
2. **Retrieval attribution is bounded, not eliminated**: the channel ablation of Section 6 shows the trajectory does not run through entity-channel fusion votes, but the full decomposition (lexicon old/new × retrieval configuration on/off) would require re-binding the collapsed

schema and re-running the collapsed-era stack—a re-collapse of a production library we did not authorize for attribution's sake.

3. **Coverage**: two knowledge bases rebuilt (61,137 + 13,408 documents) and, after the two main libraries drained (2026-10-09), the two smaller diseased libraries (1,324 + 119 documents) with the same playbook—where entity cardinality *contracted* (870→629, 209→153, with 916k/26k links and 4.6k new typed relations): the category-word aliases had inflated the count, so cardinality is not the quality metric, name distribution is. The bilingual alias layer showed isolated anomalies (an entity whose mentions vanished between extraction generations) that we document but do not fully explain.
4. **Incident-response cost**: six secondary incidents across the repair window (one silent INSERT-IGNORE swallow of seven hours, three half-open-connection wedges, a watchdog that mistook completion for failure and deleted the freshly rebuilt graph, one missing-index class) each extended the repair; we report them as findings, not hidden costs.
5. **No human-adjudicated ground truth** for entity correctness: the link-precision audit is LLM-assisted (with programmatic span verification), and entity-quality claims otherwise rest on distributional and name-sample evidence. A human-adjudicated stratified sample is the natural next evaluation step.

Data availability

The lexicon-generation tool (three-source, corpus-attested), the rebuild driver, the link-precision audit driver with all 100 adjudications (spans, verdicts, reasons), the channel-ablation runner, and the observability queries (top-10 link concentration, cardinality ratio, top-name audit, row-growth reconciliation) are released in a public artifact mirror at `https://github.com/simpleqt/knowledgenexus-artifact` (no access request needed; the institutional GitLab carries the full development history). For double-blind venues an anonymized mirror will be provided with the submission. Schema lexicons (nuclear_electronics v2.1, physics v1.1) ship as versioned assets in the same repositories. The journal corpus is third-party copyrighted; entity-layer aggregate statistics and name samples sufficient to reproduce the forensic analysis are tabulated in the released evidence pack.

Conflict of interest

The authors declare no competing interests.

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